Analysis of Lightning-Induced Fatalities in India

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1 Objective

Lightning is one of the major natural hazards in India and is responsible for a significant number of fatalities each year. Due to geographical, climatic, and socio-economic differences across states, the impact of lightning varies considerably across regions. Understanding these spatial and temporal differences is essential for assessing risk patterns and supporting informed disaster management strategies.

Although large volumes of lightning and disaster-related data are generated by governmental and meteorological agencies, such data is often published in fragmented formats and is not structured for analytical querying. Traditional operational databases are therefore inadequate for answering analytical questions such as inter-state comparisons or year-to-year variations.

In this project, a data warehouse-based framework is proposed to analyze lightning-induced fatalities in India using **state-wise annual data**. The analysis focuses on the period **2019–2020**, for which consistent lightning activity and fatality data are publicly available.

To enable deeper analytical exploration, the dataset was extended beyond the basic lightning and fatality records to include additional explanatory variables such as **population, poverty rate, literacy rate, forest cover percentage, and geographical area**. These variables provide important socio-economic and environmental context that may influence the impact of lightning across different states.

By integrating these heterogeneous datasets into a unified database structure, the project demonstrates how OLTP and data warehouse principles can be used to organize raw data for analytical processing. The resulting framework enables comparative analysis across states and years, and also establishes a structured foundation for potential future work involving statistical modeling, correlation analysis, and data mining techniques aimed at understanding the broader factors associated with lightning-induced fatalities.

2 Raw Data

The raw data used in this project was obtained from publicly available secondary sources published by official government and meteorological organizations. While these sources ensure reliability and authenticity, they differ in structure, reporting period, and level of granularity.

The primary datasets used are:

- State-wise lightning-induced fatality data obtained from official Government of India parliamentary records (Lok Sabha Unstarred Question No. 3235), compiled using statistics reported by the National Crime Records Bureau (NCRB).
- State-wise annual lightning activity data for the calendar years 2019 and 2020, obtained from officially published lightning reports and lightning detection network summaries.
- Additional socio-economic and environmental attributes including population, literacy rate, poverty rate, forest cover percentage, and geographical area obtained from publicly available government statistical reports and census-based datasets.

Although fatality data is available beyond 2020, consistent state-wise lightning activity data in calendar-year format is available only for 2019 and 2020. To maintain temporal consistency and avoid introducing assumptions, the analytical implementation of lightning activity is therefore restricted to these two years.

The inclusion of additional socio-economic and environmental variables provides broader contextual information about each state. These variables help capture factors that may influence the vulnerability of populations to lightning-related hazards. Integrating these attributes expands the analytical scope of the project and supports more meaningful exploratory analysis beyond simple lightning–fatality comparisons.

2.1 Data Quality Issues and Preprocessing Transformations

Although the data sources are authoritative, the raw datasets exhibit several quality issues commonly encountered in real-world government and meteorological data.

The major sources of noise and inconsistency identified in the raw data are listed below:

- **Formatting Noise:** Numerical attributes such as lightning counts and fatality values were reported using commas and textual formatting.
- **Structural Inconsistency:** Lightning activity data and fatality data were published as separate tables with different structures.
- **Naming Variations:** State and Union Territory names differed across sources.
- **Partial Spatial Coverage:** Certain Union Territories were absent from the lightning activity dataset.

2.2 Raw Data Representation

STATE	CLOUD-TO-GROUND	IN-CLOUD	TOTAL COUNT
Andaman and Nicobar	20,168	11,591	31,759
Andhra Pradesh	6,24,931	13,66,492	19,91,423
Arunachal Pradesh	45,240	8,521	53,761
Assam	3,24,269	3,81,084	7,05,353
Bihar	4,93,906	12,23,727	17,17,633
Chandigarh	276	268	544
Chhattisgarh	9,14,031	10,68,707	19,82,738
Dadra and Nagar Haveli	821	520	1,341
Daman and Diu	299	192	491
Goa	10,969	25,421	36,390
Gujarat	3,79,900	2,50,485	6,30,385
Haryana	76,115	91,122	1,67,237
Himachal Pradesh	65,302	32,316	97,618
Jammu and Kashmir	55,420	30,223	85,643
Jharkhand	8,54,401	15,75,011	24,29,412
Karnataka	5,62,795	15,69,092	21,31,887
Kerala	1,93,560	4,91,402	6,84,962
Lakshadweep	46	41	87
Madhya Pradesh	12,20,139	14,88,344	27,08,483
Maharashtra	7,07,769	8,72,903	15,80,672
Manipur	37,497	24,193	61,690
Meghalaya	2,15,663	4,07,715	6,23,378
Mizoram	80,283	72,454	1,52,737
Nagaland	27,452	14,410	41,862
NCT of Delhi	4,476	5,723	10,199
Odisha	16,21,408	24,63,412	40,84,820
Puducherry	1,267	3,126	4,393
Punjab	99,913	1,02,027	2,01,940
Rajasthan	5,89,299	9,77,227	15,66,526
Sikkim	4,647	2,286	6,933
Tamil Nadu	4,48,787	15,94,606	20,43,393

Figure 1: State-wise Lightning Activity for 2019

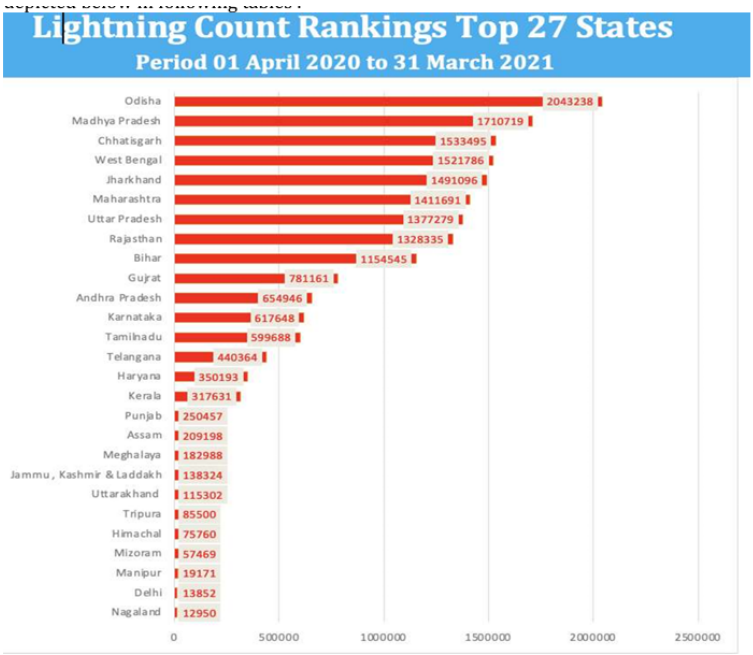

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Figure 2: State-wise Lightning Activity for 2020

Annexure-I

States/UTs	2019	2020	2021
Andhra Pradesh #	109	93	100
Arunachal Pradesh	0	0	0
Assam	28	14	16
Bihar	400	436	286
Chhattisgarh	212	246	230
Goa	2	0	1
Gujarat	64	78	40
Haryana	11	3	11
Himachal Pradesh	1	5	4
Jharkhand	334	336	254
Karnataka	99	75	121
Kerala	15	8	15
Madhya Pradesh	400	429	496
Maharashtra	189	182	238
Manipur	0	0	0
Meghalaya	7	8	7
Mizoram	1	0	0
Nagaland	0	1	0
Odisha	271	275	287
Punjab	7	8	13
Rajasthan	88	48	132
Sikkim	0	0	0
Tamil Nadu	57	64	109
Telangana #	96	64	66
Tripura	10	5	4
Uttar Pradesh	321	304	219
Uttarakhand	2	10	1
West Bengal	147	170	228
TOTAL STATE(S)	**2871**	**2862**	**2878**
A & N Islands	1	0	0
Chandigarh	2	0	0
D&N Haveli and Daman&Diu @ $	1	0	0
Delhi UT	0	0	0
Jammu & Kashmir @ *	1	0	2
Ladakh @	-	0	0
Lakshadweep	0	0	0
Puducherry	0	0	0
TOTAL UT(S)	**5**	**0**	**2**
TOTAL (ALL INDIA)	**2876**	**2862**	**2880**

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Figure 3: State-wise Lightning-Induced Fatalities

2.3 OLTP Data Pipeline

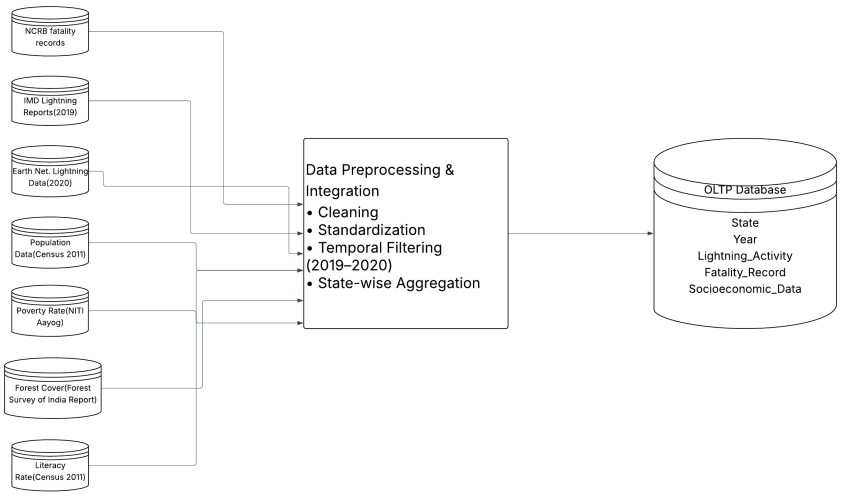

Figure 4: OLTP Data Pipeline for Lightning-Induced Fatalities Analysis

3 Data Preprocessing

Prior to integration, the datasets were examined for consistency in reporting period and spatial coverage. Only records corresponding to the calendar years **2019 and 2020** were retained for lightning activity analysis.

State and Union Territory names were standardized to resolve discrepancies arising from variations in naming conventions across different sources. Since lightning activity and fatality data originate from independent datasets, the integration was performed at an annual, state-wise aggregated level.

In addition to lightning activity and fatality records, several supplementary state-level attributes were incorporated in order to provide broader socio-economic and environmental context for the analysis. These attributes include **population, literacy rate, poverty rate, forest cover percentage, and geographical area**. These variables were obtained from publicly available government statistical reports and census-based datasets.

Unlike lightning activity and fatality records, which vary by year, these additional attributes represent relatively stable characteristics of each state. Therefore, they were treated as **static state-level variables** and linked to

the state dimension during database design.

No attempt was made to infer monthly, seasonal, or lightning sub-type information, as such details were not uniformly available. This preprocessing approach prioritizes data integrity and avoids the introduction of unsupported assumptions.

3.1 Preprocessed Data Representation

After applying the preprocessing steps, the cleaned datasets were organized into structured tabular formats suitable for database storage. For clarity and compact presentation, representative subsets of the preprocessed datasets are shown below.

Table 1: Sample of Preprocessed State-wise Lightning-Induced Fatalities (2019–2020)

State	**Year**	**Deaths**
Andhra Pradesh	2019	109
Andhra Pradesh	2020	93
Assam	2019	28
Assam	2020	14
Bihar	2019	400
Bihar	2020	436
Goa	2019	2
Goa	2020	0
Gujarat	2019	64
Gujarat	2020	78

Table 2: Sample of Preprocessed State-wise Lightning Intensity (2019–2020)

State	**Year**	**Lightning Intensity**
Andhra Pradesh	2019	1,991,423
Andhra Pradesh	2020	3,919,947
Assam	2019	705,353
Assam	2020	339,664
Bihar	2019	1,717,633
Bihar	2020	1,525,765
Goa	2019	36,390
Goa	2020	15,883
Gujarat	2019	630,385
Gujarat	2020	1,090,153

Table 3: Sample of State-Level Socioeconomic and Environmental Variables

State	Population	Area (km^2)	Literacy (%)	Forest (%)	Poverty (%)
Andaman & Nicobar Islands	380581	8249	86.63	94.36	1.63
Andhra Pradesh	49577103	162975	67.02	22.63	4.19
Arunachal Pradesh	1383727	83743	65.38	92.86	10.05
Assam	31205576	78438	72.19	23.36	14.47
Bihar	104099452	94163	61.80	6.64	26.59
Chandigarh	1055450	114	86.05	25.16	2.62
Chhattisgarh	25545198	135192	70.28	46.89	11.71
Dadra & Nagar Haveli and Daman & Diu	586620	603	76.34	42.91	6.06

4 OLTP Design

4.1 Normalized OLTP Schema

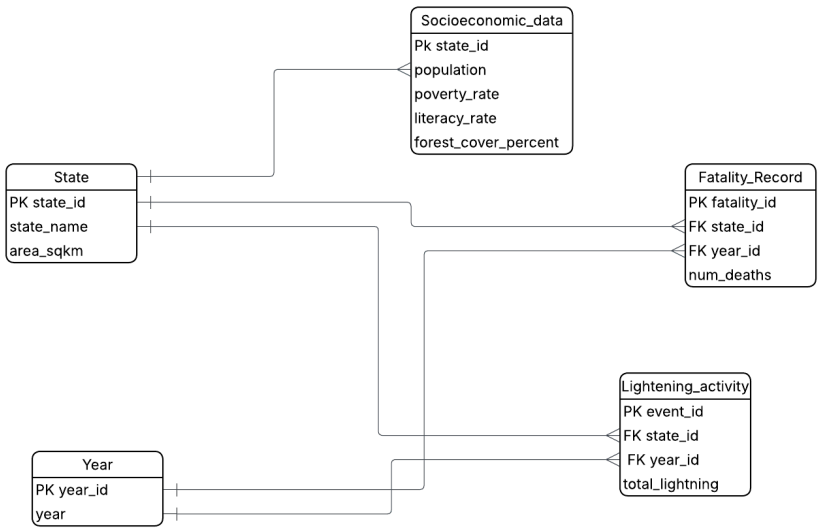

Figure 5: Normalized OLTP Schema for Lightning Data

5 Data Warehouse Design

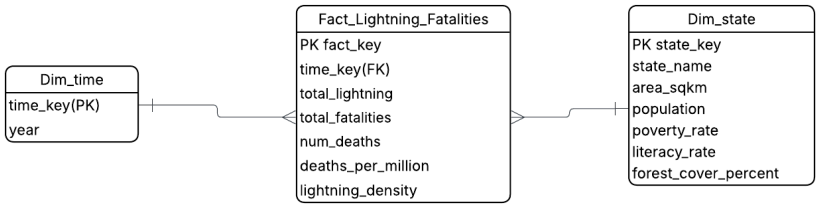

Figure 6: Star Schema for Lightning-Induced Fatalities Data Warehouse

6 OLAP Queries and Analytical Processing

The following OLAP queries were executed on the populated data warehouse to analyze spatial and temporal patterns in lightning-induced fatalities across Indian states. Each query result is accompanied by a brief interpretation based on the observed output.

6.1 State-wise Lightning Intensity and Fatalities

```
SELECT DISTINCT
       s.state_name,
       f.year,
       f.deaths,
       l.intensity_value
FROM states s
JOIN fatalities f ON s.state_id = f.state_id
JOIN lightning_intensity l
     ON f.state_id = l.state_id
    AND f.year = l.year
WHERE f.year IN (2019, 2020);
```

state_name	year	deaths	intensity_value
andhra pradesh	2019	109	1991420
andhra pradesh	2020	93	3919950
arunachal pradesh	2019	0	53761
arunachal pradesh	2020	0	26448
assam	2019	28	705353
assam	2020	14	339664
bihar	2019	400	1717630
bihar	2020	436	1525760
chhattisgarh	2019	212	1982740
chhattisgarh	2020	246	2708930
goa	2019	2	36390
goa	2020	0	15883
gujarat	2019	64	630385

Figure 7: State-wise Lightning Intensity and Fatalities at Annual Granularity (2019–2020)

This query integrates lightning intensity and fatality data at a common state–year level. The results provide a foundational view for comparing lightning activity with associated human impact across regions and years.

—

6.2 States with Highest Average Lightning Intensity

```
SELECT s.state_name, AVG(l.intensity_value) AS avg_intensity
FROM state s
JOIN lightning_intensity l ON s.state_id = l.state_id
GROUP BY s.state_name
ORDER BY avg_intensity DESC;
```

state_name	avg_intensity
tamil nadu	3987904
odisha	3890044.5
west bengal	3024659.5
andhra pradesh	2955685
karnataka	2594975
madhya pradesh	2548656
jharkhand	2371075.5
chhattisgarh	2345833
maharashtra	2017917
uttar pradesh	1829644
bihar	1621699
telangana	1500080
rajasthan	1186295
kerala	1100039
gujarat	860269
assam	522508.5
meghalaya	491544.5
tripura	182346.5
haryana	172298.5
mizoram	109528
punjab	106822

Figure 8: Average Lightning Intensity by State (2019–2020)

The results highlight states that consistently experience higher lightning activity over the study period, revealing clear spatial variation in lightning intensity across India.

—

6.3 States with Highest Total Fatalities

```
SELECT s.state_name, SUM(f.deaths) AS total_deaths
FROM state s
JOIN fatalities f ON s.state_id = f.state_id
GROUP BY s.state_name
ORDER BY total_deaths DESC;
```

state_name	total_deaths
bihar	1672
madhya pradesh	1658
jharkhand	1340
uttar pradesh	1250
odisha	1092
chhattisgarh	916
maharashtra	742
west bengal	634
andhra pradesh	404
karnataka	348
telangana	320

Figure 9: Total Lightning-Induced Fatalities by State (2019–2020)

This query identifies states with the highest cumulative lightning-induced fatalities. The output indicates that states with high fatality counts are not necessarily those with the highest lightning intensity.

—

6.4 Year-wise Fatality Trend

```
SELECT year, SUM(deaths) AS total_deaths
FROM fatalities
GROUP BY year
ORDER BY year;
```

```
    -> ORDER BY year;
+------+-----------------+
| year | deaths_per_year |
+------+-----------------+
| 2019 |            5762 |
| 2020 |            5724 |
+------+-----------------+
```

Figure 10: Year-wise Lightning-Induced Fatalities in India (2019–2020)

The year-wise aggregation enables temporal comparison of fatalities. The observed variation reflects changes in lightning impact across the two years under study.

—

6.5 High Intensity but Low Fatality States

```
SELECT s.state_name,
       AVG(l.intensity_value) AS avg_intensity,
       AVG(f.deaths) AS avg_deaths
FROM state s
JOIN lightning_intensity l ON s.state_id = l.state_id
JOIN fatalities f ON s.state_id = f.state_id
AND l.year = f.year
GROUP BY s.state_name
HAVING AVG(l.intensity_value) >
       (SELECT AVG(intensity_value) FROM lightning_intensity)
AND AVG(f.deaths) <
       (SELECT AVG(deaths) FROM fatalities);
```

```
    -> AND avg_deaths <
    ->        (SELECT AVG(deaths) FROM fatalities);
+------------+---------------+------------+
| state_name | avg_intensity | avg_deaths |
+------------+---------------+------------+
| kerala     |       1100039 |    11.5000 |
| rajasthan  |       1186295 |    68.0000 |
| tamil nadu |       3987904 |    60.5000 |
| telangana  |       1500080 |    80.0000 |
+------------+---------------+------------+
```

Figure 11: States with High Lightning Intensity but Low Fatality Rates

This query reveals states where high lightning activity does not proportionally translate into high fatalities. Such patterns suggest the influence of factors such as early warning systems, public awareness, and infrastructure resilience.

6.6 States with Highest Lightning Fatalities per Million Population

```
SELECT s.state_name,
       sm.year,
       sm.deaths_per_million
FROM state_metrics sm
JOIN state s ON sm.state_id = s.state_id
ORDER BY sm.deaths_per_million DESC;
```

state_name	year	deaths_per_million
jharkhand	2020	10.1855
jharkhand	2019	10.1249
chhattisgarh	2020	9.62999
chhattisgarh	2019	8.29902
odisha	2020	6.55164
odisha	2019	6.45634
madhya pradesh	2020	5.90691
madhya pradesh	2019	5.50761
bihar	2020	4.1883
puducherry	2019	4.00656
bihar	2019	3.84248
telangana	2019	2.74257
tripura	2019	2.72189
meghalaya	2020	2.69643
a & n islands	2019	2.62756
meghalaya	2019	2.35937
andhra pradesh	2019	2.1986
chandigarh	2019	1.89493
andhra pradesh	2020	1.87587
west bengal	2020	1.86248

Figure 12: Lightning Fatalities per Million Population by State (2019–2020)

This query retrieves lightning fatalities normalized by population size using the deaths-per-million metric. Such normalization enables meaningful comparison of lightning risk across states with varying population levels.

6.7 States with Highest Lightning Density

```
SELECT s.state_name,
       sm.year,
       sm.lightning_density
FROM state_metrics sm
JOIN state s ON sm.state_id = s.state_id
ORDER BY sm.lightning_density DESC;
```

state_name	year	lightning_density
puducherry	2020	48.952
tamil nadu	2020	45.6136
kerala	2020	38.9861
west bengal	2020	34.3669
west bengal	2019	33.7929
jharkhand	2019	30.4758
jharkhand	2020	29.0122
meghalaya	2019	27.7934
odisha	2019	26.234
andhra pradesh	2020	24.0524
odisha	2020	23.7322
tripura	2019	20.3312
chhattisgarh	2020	20.0376
bihar	2019	18.2411
kerala	2019	17.625
telangana	2020	16.7968
bihar	2020	16.2034
meghalaya	2020	16.0378

Figure 13: Lightning Density by State and Year (2019–2020)

Lightning density measures the concentration of lightning events per square kilometer. States with higher values experience more intense lightning activity relative to their geographical area.

6.8 Fatalities with Socioeconomic Context

```
SELECT s.state_name,
       SUM(f.deaths) AS total_deaths,
       sv.poverty_rate,
       sv.literacy_rate,
       sv.forest_cover
FROM fatalities f
JOIN state s ON f.state_id = s.state_id
JOIN static_variables sv ON s.state_id = sv.state_id
GROUP BY s.state_name, sv.poverty_rate, sv.literacy_rate, sv.forest_cover
ORDER BY total_deaths DESC;
```

state_name	total_deaths	poverty_rate	literacy_rate	forest_cover
bihar	836	26.59	61.8	6.64
madhya pradesh	829	15.01	69.32	28.31
jharkhand	670	23.34	66.41	28.1
uttar pradesh	625	17.4	67.68	7.54
odisha	546	11.07	72.87	35.13
chhattisgarh	458	11.71	70.28	46.89
maharashtra	371	5.48	82.34	16.94
west bengal	317	8.6	76.36	13.53
andhra pradesh	202	4.19	67.02	22.63
karnataka	174	5.67	75.36	16.13
telangana	160	3.76	66.46	24.7
gujarat	142	9.03	78.03	10.91
rajasthan	136	10.77	66.11	8.1
tamil nadu	121	1.43	80.09	16.54
assam	42	14.47	72.19	23.36
kerala	23	0.48	94	27.83

Figure 14: Lightning-Induced Fatalities with Socioeconomic Indicators by State

This query compares lightning-induced fatalities with socioeconomic indicators such as poverty rate, literacy rate, and forest cover. Such contextual analysis helps explore possible factors influencing lightning vulnerability across different states.

6.9 Combined View of Lightning Intensity and Fatalities

```
SELECT s.state_name,
       f.year,
       l.intensity_value,
       f.deaths
FROM fatalities f
JOIN lightning_intensity l
     ON f.state_id = l.state_id
    AND f.year = l.year
JOIN state s
     ON f.state_id = s.state_id
ORDER BY f.year, f.deaths DESC;
```

state_name	year	intensity_value	deaths
bihar	2019	1717630	400
madhya pradesh	2019	2708480	400
jharkhand	2019	2429410	334
uttar pradesh	2019	1948020	321
odisha	2019	4084820	271
chhattisgarh	2019	1982740	212
maharashtra	2019	1580670	189
west bengal	2019	2999190	147
andhra pradesh	2019	1991420	109
karnataka	2019	2131890	99
telangana	2019	1117630	96
rajasthan	2019	1566530	88
gujarat	2019	630385	64
tamil nadu	2019	2043390	57
assam	2019	705353	28
kerala	2019	684962	15
haryana	2019	167237	11

Figure 15: Combined View of Lightning Intensity and Fatalities by State (2019-2020)

This query presents lightning intensity and lightning-induced fatalities together for each state and year. By combining both measures in a single analytical view, the results enable direct comparison between lightning activity levels and their associated human impact. Such comparisons help identify states where high lightning activity corresponds to high fatality counts, as well as cases where strong lightning activity does not translate into proportional fatalities.

7 Graphical Analysis and Visualization

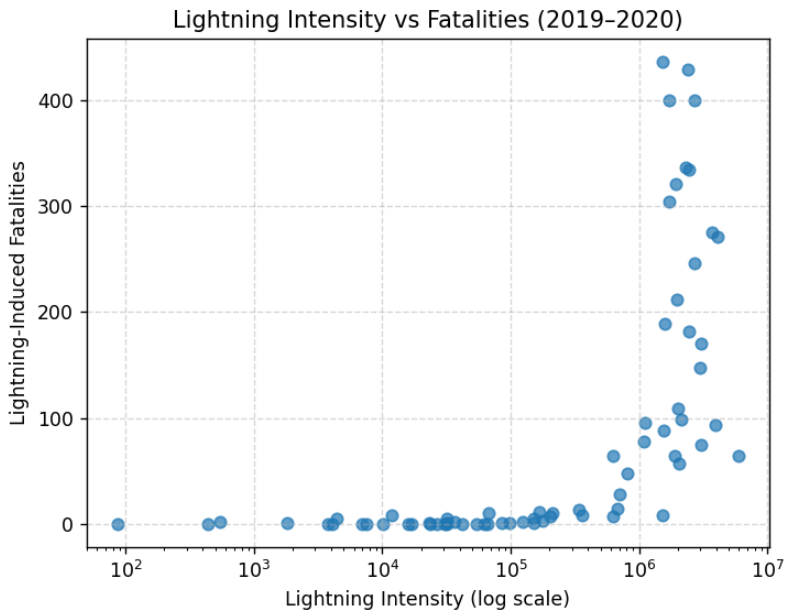

Figure 16: Lightning Intensity vs Lightning-Induced Fatalities (2019–2020)

The scatter plot illustrates the relationship between lightning intensity and lightning-induced fatalities at the state–year level. The log-scaled intensity axis reveals a dispersed pattern, indicating that higher lightning activity does not consistently correspond to higher fatality counts. This suggests that factors such as preparedness, awareness, and infrastructure may play a significant role in mitigating lightning-related fatalities.

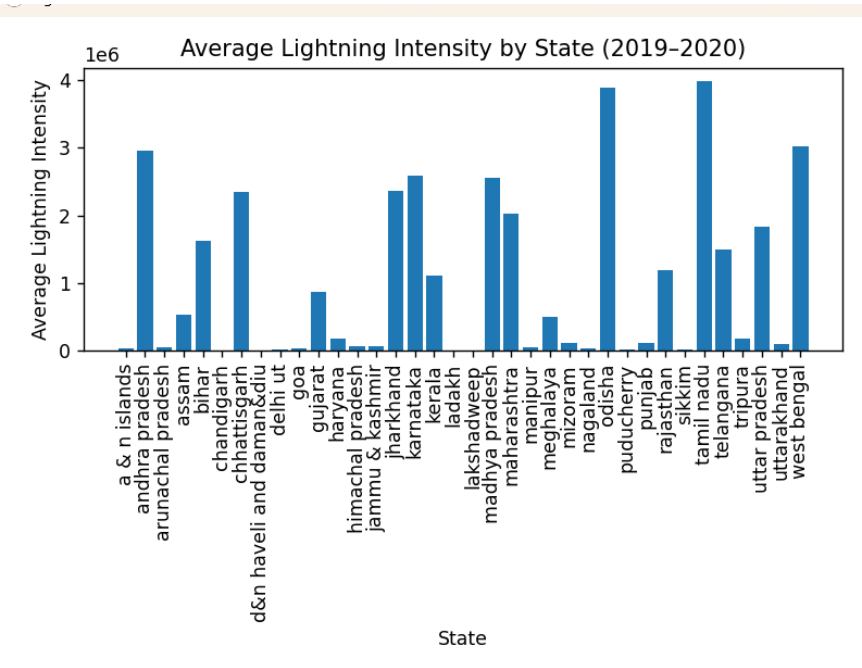

Figure 17: Average Lightning Intensity by State (2019–2020)

The state-wise bar chart highlights substantial spatial variation in lightning activity across India. Comparison with fatality statistics indicates that states experiencing higher lightning intensity are not necessarily those with the highest cumulative fatalities, reinforcing observations from the OLAP analysis.

While the analysis highlights instances where high lightning intensity coincides with comparatively low fatality counts, it is important to note that the available data does not allow for direct attribution of such patterns to specific policy interventions or mitigation measures. The observed relationships are descriptive in nature and may also be influenced by unobserved factors such as population distribution, occupational exposure, reporting practices, and regional geography. Consequently, causal interpretations are beyond the scope of the present study.

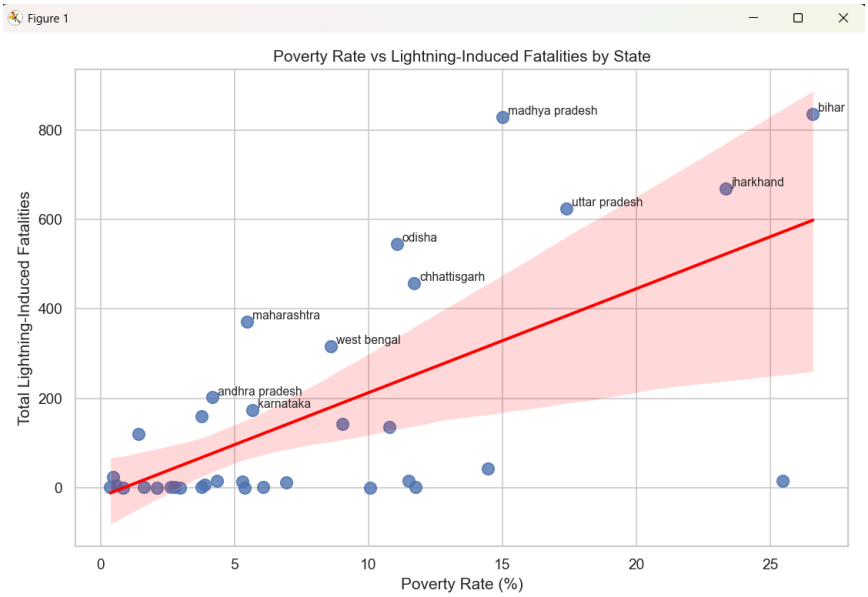

Figure 18: Relationship between Poverty Rate and Lightning-Induced Fatalities by State

To further explore potential socioeconomic influences on lightning fatalities, a scatter plot was generated comparing poverty rates with total lightning-induced fatalities across states. The visualization suggests that states with higher poverty rates tend to report relatively higher fatality counts. Although the relationship is not strictly linear, the observed pattern indicates that socioeconomic vulnerability may contribute to increased exposure to lightning-related hazards.

Higher poverty levels may correlate with factors such as outdoor occupations, limited access to safe shelter during storms, and lower access to early warning systems. These conditions can increase the likelihood of lightning-related injuries and fatalities. However, it is important to note that this relationship is exploratory and descriptive in nature, and other factors such as population size, geographical conditions, and reporting practices may also influence the observed patterns.

8 Summary and Scope of the Project

This project presented a data warehouse–based framework for analyzing lightning-induced fatalities in India using state-wise annual data. The primary objective was to demonstrate how heterogeneous real-world datasets, obtained from multiple governmental and statistical sources, can be systematically integrated and organized to support analytical querying and exploratory analysis.

The raw datasets included lightning activity statistics, lightning-induced fatality records, and additional state-level socioeconomic and environmental attributes such as population, geographical area, literacy rate, forest cover percentage, and poverty rate. These datasets were collected from publicly available government reports and statistical repositories. Since the sources differed in format and structure, a series of preprocessing steps were performed including data cleaning, standardization of state names, and temporal filtering to ensure consistency across datasets.

An ELT-based data pipeline was implemented in which the cleaned datasets were first loaded into staging tables and subsequently transformed into a normalized OLTP schema. The OLTP schema organized the data into relational tables representing states, lightning activity, fatality records, and static socioeconomic variables. Derived analytical attributes such as lightning density and fatalities per million population were then computed to enrich the analytical capability of the system.

To support multidimensional analysis, the normalized OLTP schema was further transformed into a star-based data warehouse model consisting of fact and dimension tables. This design enabled efficient OLAP queries to examine spatial and temporal patterns in lightning activity and associated fatalities across Indian states.

Using the implemented data warehouse, several OLAP queries were executed to analyze relationships between lightning intensity, fatalities, and socioeconomic indicators. Additional graphical visualizations were generated to explore potential associations between lightning-related fatalities and factors such as lightning intensity and poverty levels. The analysis

highlights significant regional variation and suggests that lightning intensity alone does not fully explain fatality patterns, indicating the potential influence of socioeconomic vulnerability and other contextual factors.

The scope of this project is intentionally constrained by the temporal coverage and granularity of the available data, which currently includes state-level annual observations for the period 2019–2020. Future work may involve extending the dataset across additional years, incorporating higher temporal resolution data such as seasonal or monthly lightning activity, and applying statistical or machine learning techniques to support predictive modeling and risk assessment.

9 Machine Learning and Pattern Mining

Building upon the OLAP-based exploratory analysis presented in the previous sections, machine learning techniques were applied to further investigate the factors influencing lightning-induced fatalities across Indian states. The objective of this phase is to move beyond descriptive analysis and identify underlying patterns, relationships, and predictive insights within the data.

9.1 Problem Formulation

The task is formulated as a supervised regression problem, where the target variable is the number of lightning-induced fatalities at the state-year level. The objective is to model fatalities as a function of environmental and socioeconomic variables.

The input features considered in the analysis include:

- Lightning intensity
- Poverty rate
- Literacy rate
- Population
- Forest cover percentage

- Lightning density

These features were selected based on their relevance to exposure, vulnerability, and environmental conditions identified during the data warehouse analysis.

9.2 Data Preparation

The analytical dataset was constructed by integrating multiple tables from the data warehouse, including fatalities, lightning intensity, and state-level socio-economic attributes. SQL joins were used to extract a unified dataset at the state-year granularity.

Prior to model training, the dataset was preprocessed to remove missing values and ensure consistency across all variables. Only numerical attributes were retained for machine learning analysis.

9.3 Regression Analysis

A multiple linear regression model was employed as a baseline approach to quantify the relationship between fatalities and the selected features. This model provides interpretable coefficients that indicate the direction and magnitude of influence of each variable.

Regression analysis enables an understanding of how changes in socio-economic and environmental factors impact fatality outcomes, thereby supporting the interpretability of the results.

9.4 Decision Tree Model

To capture potential non-linear relationships and interactions between variables, a decision tree regressor was implemented. Unlike linear regression, decision trees partition the data based on hierarchical decision rules, allowing for more flexible modeling of complex patterns.

The decision tree structure provides intuitive insights into how different variables contribute to fatality outcomes. The hierarchical splits reveal the relative importance of features and highlight threshold-based effects.

9.5 Feature Importance Analysis

Feature importance scores were derived from the decision tree model to quantify the contribution of each variable to the prediction of fatalities. This analysis helps identify the most influential factors driving lightning-induced fatalities.

Understanding feature importance is critical for interpreting model behavior and aligning the results with the broader objective of assessing vulnerability and risk.

9.6 Clustering Analysis

In addition to supervised learning, unsupervised learning techniques were applied to identify patterns within the data. K-means clustering was used to group states into distinct clusters based on their feature profiles.

This approach enables the identification of states with similar characteristics in terms of lightning exposure and socio-economic conditions. The resulting clusters provide insights into varying levels of vulnerability and risk across regions.

9.7 Summary

The combination of regression, decision tree modeling, feature importance analysis, and clustering provides a comprehensive analytical framework. These techniques complement the earlier OLAP-based analysis and enable deeper understanding of the factors influencing lightning-induced fatalities.

Together, these methods support both interpretability and pattern discovery, forming a strong foundation for data-driven insights in the context of disaster risk analysis.

10 Results and Insights

The application of machine learning techniques provides deeper insights into the factors influencing lightning-induced fatalities across Indian states. This

section presents the key findings derived from correlation analysis, regression modeling, decision tree analysis, and clustering.

10.1 Correlation Analysis

A correlation matrix was computed to examine the relationships between fatalities and the selected features. The analysis reveals that lightning intensity exhibits a moderate positive correlation with fatalities, indicating that higher lightning activity contributes to increased fatality counts.

However, socio-economic variables also show significant relationships. In particular, poverty rate demonstrates a strong positive correlation with fatalities, suggesting that economically vulnerable populations are more susceptible to lightning-related hazards. Conversely, literacy rate exhibits a negative correlation with fatalities, indicating that higher levels of education and awareness may reduce the risk of fatal outcomes.

Population also shows a notable positive correlation with fatalities, reflecting increased exposure in densely populated regions. These findings highlight that lightning-induced fatalities are influenced by a combination of environmental exposure and socio-economic vulnerability.

10.2 Regression Insights

The multiple linear regression model provides further evidence supporting the observed relationships. The model coefficients indicate that lightning intensity and poverty rate contribute positively to fatalities, while literacy rate has a negative impact.

These results reinforce the hypothesis that while environmental factors such as lightning intensity are important, socio-economic conditions play a crucial role in determining the severity of outcomes. The regression model offers an interpretable framework for understanding how different variables contribute to fatality levels.

10.3 Decision Tree Analysis

The decision tree model captures non-linear relationships and hierarchical interactions between variables. The tree structure reveals that lightning intensity serves as the primary splitting criterion, indicating its dominant role in determining fatality levels.

Subsequent splits based on poverty rate and population highlight the importance of socio-economic conditions in influencing vulnerability. The hierarchical nature of the model demonstrates that fatalities are not determined by a single factor, but rather by a combination of interacting variables.

10.4 Feature Importance

Feature importance analysis derived from the decision tree model indicates that lightning intensity is the most influential variable, followed by poverty rate. Population contributes to a lesser extent, while variables such as forest cover and lightning density have relatively lower impact.

These results emphasize that while environmental exposure is a key driver, socio-economic vulnerability significantly affects the observed fatality patterns.

10.5 Clustering and Risk Patterns

K-means clustering was applied to group states based on their feature profiles. The clustering results reveal the presence of distinct groups of states with varying levels of risk.

Some clusters correspond to states with high lightning intensity but relatively lower fatalities, suggesting better preparedness or mitigation measures. Other clusters represent states with higher fatalities despite moderate lightning intensity, indicating increased vulnerability due to socio-economic factors.

This pattern-based analysis highlights that risk is not uniform across regions and is influenced by both environmental and socio-economic dimensions.

10.6 Overall Interpretation

The combined results from correlation analysis, regression modeling, decision tree analysis, and clustering provide a consistent and coherent understanding of the problem. The findings indicate that lightning intensity alone does not fully explain fatality patterns.

Instead, socio-economic variables such as poverty and literacy significantly influence vulnerability, while population determines the level of exposure. These insights underscore the importance of integrating environmental and socio-economic factors in assessing disaster risk.

Overall, the analysis demonstrates that lightning-induced fatalities are driven by a complex interplay of multiple factors, rather than a single dominant variable.

11 Model Evaluation and Visualization

To support the analytical findings, several visualizations were generated to evaluate model behavior and illustrate key patterns identified through machine learning techniques.

11.1 Correlation Heatmap

A correlation heatmap was used to visualize the relationships between fatalities and the selected features. The heatmap highlights both positive and negative correlations, providing an overview of how environmental and socio-economic variables interact with fatality outcomes.

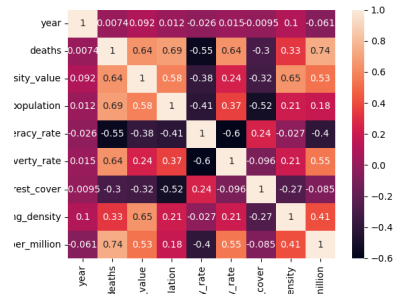

Figure 19: Correlation Heatmap of Selected Features

11.2 Regression Fit

To evaluate the performance of the regression model, a scatter plot of actual versus predicted fatalities was generated. The plot provides a visual assessment of how well the model captures the observed data patterns.

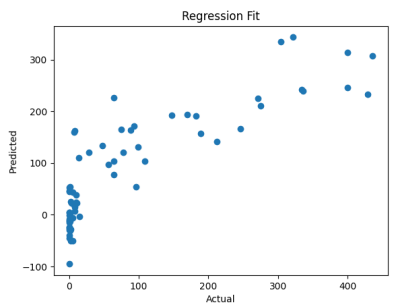

Figure 20: Actual vs Predicted Fatalities

11.3 Decision Tree Visualization

The decision tree model was visualized to illustrate the hierarchical structure of feature-based splits. The visualization provides an interpretable representation of how different variables contribute to predicting fatalities.

Figure 21: Decision Tree Representation of Fatality Prediction

11.4 Feature Importance

A feature importance plot was generated to quantify the contribution of each variable to the prediction task. This visualization highlights the relative importance of environmental and socio-economic factors.

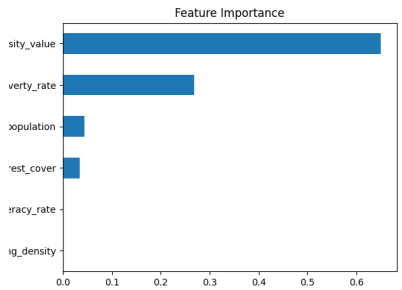

Figure 22: Feature Importance for Fatality Prediction

11.5 Clustering Analysis and Visualization

K-means clustering was used to group states based on their feature profiles. The interpretation of clusters was performed using the average values of features within each cluster, which provides meaningful insights into the characteristics of each group.

For visualization purposes, Principal Component Analysis (PCA) was applied to project the high-dimensional feature space into two dimensions. This projection helps illustrate the separation between clusters, although the axes do not directly correspond to original variables.

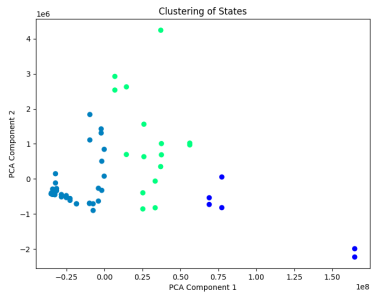

Figure 23: Visualization of State Clusters using PCA Projection

11.6 Summary

The visualizations support the findings from the machine learning models by providing intuitive representations of relationships and patterns in the data. While PCA aids in visualizing cluster separation, the interpretation of clusters is based on feature statistics, ensuring meaningful and accurate insights.

12 Scalability, Performance and Limitations

This section discusses the performance characteristics of the system, along with its scalability considerations, memory and computational efficiency, and inherent limitations.

12.1 System Performance

The data warehouse was implemented using a relational database system, enabling efficient execution of OLAP queries for analytical processing. The use of structured schemas, normalized OLTP design, and star schema transformation ensured optimized query execution through indexed joins and reduced redundancy.

Given the relatively small dataset (state-level annual data), query response times were low, and aggregation operations such as GROUP BY and JOIN were executed efficiently. The ELT pipeline design further ensured that preprocessing overhead was handled prior to analytical querying, improving runtime performance during OLAP operations.

The machine learning models applied in this study, including linear regression and decision tree regression, are computationally lightweight. As a result, both training and inference were performed efficiently on standard hardware without requiring specialized computational resources such as GPUs.

12.2 Memory and Storage Considerations

The current system operates within modest memory and storage constraints due to the limited dataset size. However, as the data warehouse grows, memory usage may increase due to:

- Storage of large fact tables containing multi-year, high-resolution data
- Intermediate query results during OLAP operations
- In-memory data structures used during machine learning model training

In large-scale scenarios, this may lead to increased I/O overhead and memory bottlenecks. Techniques such as data compression, columnar storage formats, and efficient indexing mechanisms can help mitigate these challenges.

12.3 Scalability Considerations

While the current implementation operates on a relatively small dataset, the architecture is designed to be extensible. The data warehouse framework supports integration of additional datasets, higher temporal resolution (monthly/daily), and finer spatial granularity (district-level data).

However, as data volume and dimensionality increase, several scalability challenges may arise:

- Increased query execution time due to complex multi-table joins
- Higher storage requirements for large fact tables
- Slower ETL/ELT processing pipelines
- Increased training time for machine learning models

To address these challenges, future implementations may incorporate:

- Distributed data processing frameworks such as Apache Spark

- Data partitioning (e.g., by year or region) to improve query efficiency
- Parallel query execution and indexing strategies
- Use of data lakes for handling raw large-scale datasets

12.4 Compute Efficiency

The current system is computationally efficient due to:

- Use of simple regression and tree-based models
- Limited feature space and dataset size
- Pre-aggregation of data in the warehouse

However, as more complex models (e.g., ensemble methods or deep learning models) are introduced, computational costs may increase significantly. Training such models may require:

- Hardware acceleration (GPUs/TPUs)
- Batch processing or mini-batch optimization
- Efficient feature engineering pipelines

12.5 Pipeline and System-Level Limitations

The ELT pipeline used in this project performs data cleaning, transformation, and integration in a sequential manner. While suitable for small datasets, this approach may become a bottleneck when handling large-scale or real-time data.

Potential limitations include:

- Lack of real-time data ingestion capability
- Sequential processing limiting throughput
- Manual preprocessing steps reducing automation

Future systems could adopt streaming pipelines (e.g., Kafka-based architectures) and automated workflows to improve scalability and robustness.

12.6 Limitations of the Study

The primary limitation of this study arises from the limited temporal and spatial granularity of the available data. The analysis is restricted to state-level annual data for the period 2019–2020, which constrains the ability to capture seasonal variations and localized patterns.

Additionally, the relatively small dataset limits the complexity and predictive power of machine learning models. The results obtained are exploratory in nature and should not be interpreted as definitive causal relationships.

Furthermore, several potentially important variables were not included due to data unavailability, including:

- High-resolution meteorological variables (e.g., rainfall, temperature, storm patterns)
- Real-time lightning detection data
- Infrastructure and early warning system availability
- Occupational exposure patterns

12.7 Summary

Overall, the system demonstrates strong performance and efficiency for small-scale analytical workloads. However, scaling to larger datasets and real-time applications will require distributed architectures, improved data pipelines, and more advanced computational resources.

12.8 Future Work

Future work may focus on expanding the dataset across multiple years and incorporating higher-resolution temporal data, such as monthly or seasonal observations. Integration of additional variables, including meteorological parameters and infrastructure-related indicators, may further enhance the analysis.

Advanced machine learning models, including ensemble methods and deep learning approaches, may also be explored to improve predictive accuracy. Additionally, real-time data integration and interactive dashboards could support practical decision-making in disaster management.

12.9 Summary

Despite the limitations, the current framework provides a scalable and extensible foundation for analyzing lightning-induced fatalities. The integration of data warehousing and machine learning techniques demonstrates the potential for data-driven approaches in understanding and mitigating disaster risks.

13 Conclusion

This study presents a comprehensive analysis of lightning-induced fatalities in India by integrating data warehousing techniques with machine learning methodologies. The project begins with the construction of a structured data warehouse, enabling efficient storage, querying, and analytical processing of multi-source data.

OLAP-based analysis provides initial insights into regional variations in lightning intensity and fatalities, highlighting the importance of both environmental and socio-economic factors. Building upon this foundation, machine learning techniques were applied to uncover deeper patterns and relationships within the data.

The results demonstrate that while lightning intensity is a significant contributing factor, socio-economic variables such as poverty and literacy play a crucial role in determining vulnerability. Decision tree analysis further reveals the hierarchical influence of these factors, while clustering identifies distinct groups of states with varying risk profiles.

Additionally, the development of an interactive Gradio-based interface transforms the analytical model into a user-friendly system, allowing real-time exploration of how different parameters influence fatality predictions.

This enhances the practical applicability of the project and bridges the gap between data analysis and decision support.

Overall, the integration of data warehousing, machine learning, and interactive visualization provides a robust framework for understanding disaster-related risks. The study highlights the importance of combining environmental and socio-economic perspectives in developing effective mitigation strategies. Future work may extend this framework with larger datasets, real-time data integration, and advanced predictive models to further improve accuracy and applicability.